

Sensor Configuration and Data Transfer from Instruments

Van Essen CTD-Diver

Notes:

- For the first time data is transferred from the instrument, make sure to download and install the Diver-Office software from [Diver-Office - Van Essen Instruments](#)
 - The software only functions with Windows 10 or 11 operating systems.
- A Van Essen Diver USB reading unit is needed to connect the instrument to a USB port on a computer and is supplied with the instrument

Configuring Settings for the Instrument:

1. When the Diver-Office software is first opened, the user will be prompted to create a project. The project folder path will be shown and the user can select where the folder will be located on the host computer.
 - a. Users can define which units (found in preferences) that the CTD-Diver will report in and the language:
 - i. Pressure = mH₂O
 - ii. Temperature = Celsius
 - iii. Conductivity = mS/cm
 - iv. Altitude = meter
 - b. The 'Default Export Settings' should be edited to export the file as a .DAT, .MON and .CSV; keep the default filename format the same.
 - c. Users can choose the 'Project Time Settings' from a list of options that will sync to the current computer time, standard time (winter), daylight saving time (summer), or coordinated universal time (UTC).
 - i. We recommend UTC time.
 - d. The 'Vertical Reference Datum' should be selected for "Mean Sea Level (MSL)".
 - e. Keep the 'Merge Time Series' selected as "manual".
2. Unscrew the cap protecting the optical port on the CTD-Diver and insert the instrument into the base of the USB reading unit and connect the USB reader to the USB port of a computer.
 - a. If there are issues connecting the CTD-Diver to the Diver-Office software, see the troubleshooting sheet that came with the USB reading unit.
3. Click the "Diver" button to open the Diver window.
4. To set the Diver recording settings, enter information including:
 - a. A name for the study for the monitoring point name, or can be left as the serial number of the unit.
 - b. "Fixed-length memory" for the sample method.

- c. The desired recording interval. Suggested recording interval for GEM-in-a-Box is 1 minute.
 - d. "120 mS/cm" for the conductivity range.
 - e. "1: Cond." for the conductivity type.
5. Click the "Program" button to save the settings.
6. Click the "Start" button to open the Start Diver Dialog.
7. Select the desired start method for sampling:
 - a. "Immediate Start" : the Diver starts immediately collecting data.
 - i. Click this option if sampling will begin within 15 minutes after leaving the lab.
 - b. "Future Start" : the Diver starts collecting data at a user specified future start time (YYYY-MM-DD HH-MM-SS AM/PM).
 - i. Click this option and set an approximate time that you will begin sampling, always err on the side of starting before the exact time you will arrive/begin.
 - c. Click to checkmark the box to "sync diver time with project time".
8. Click the "Start" button to save the settings to the CTD-Diver.
9. When completed, click "Close" on the Start Diver Dialog box.
10. Disconnect the CTD-Diver from the USB reader. The CTD-Diver is now ready to be used in the field for water sampling.

Reading Data from the Instrument:

11. Dry the housing of the CTD-Diver and remove the end cap covering the optical data port.
12. Attach the CTD-Diver onto the USB reader connected to the computer and open the Diver-Office software. Connect to the CTD-Diver and click the "Stop" button.
13. Click the "Data" button to download data from the CTD-Diver. Clicking the down arrow next to the "Data" button enables to change the mode/type of data download between:
 - a. "Data" : download all data recorded on the instrument.
 - b. "New Data" : download only newly recorded data since the last data download.
 - c. "Backup Data" : download data from the previous monitoring session.
14. The download progress will be indicated by a progress bar and once complete, the data can be exported automatically if this option is selected in the Project Settings, or can be manually exported by right-clicking the dataset file and selecting 'Export'
 - a. Edit the 'Export file type(s)' to export the file as a .CSV.
 - b. Keep the default filename format for the download.

15. After downloading the data, Diver-Office will then open a view displaying the downloaded time series and a graph/table of the data.
16. When completed, click “Close” on the Dialog box. Disconnect from the CTD-Diver in the software, then disconnect the USB reader unit from the computer and the instrument.
17. Upload raw files to the Github repository and complete an Instrument Data Processing sheet.

PME miniDOT Logger

Notes:

- The miniDOT programs require Java 1.7 or higher installed on the computer.
- For the first time data is transferred from the instrument, after connecting the miniDOT logger to the computer USB port, copy and paste the miniDOTControl.jar, miniDOTPlot.jar, and miniDOTConcatenate.jar programs onto the computer to install and run the programs for data transfer and analyses. They can also be run from the logger itself. Also transfer the minidotAdjustCalibration.jar from the computer to the logger if not already onboard the unit.
- The miniDOT logger records measurements of time (whichever time zone the host computer is reporting in), battery voltage, temperature (°C), dissolved oxygen (mg/L - not compensated for salinity or pressure), and a dimensionless parameter Q, an indicator of the quality of the optical measurement.
- The time and temperature data can be used to match the DO measurement to the corresponding depth on the Van Essen CTD-Diver (records time, temperature, depth, conductivity).
- The recording time interval can be adjusted using the miniDOT Control program



Figure 1. Layout of the miniDOT control interface

Sensor Configuration

1. Open the miniDOT logger by unscrewing the white pressure housing from the black end cap. Remove the white pressure housing completely, being careful not to drip water onto the interior surfaces of circuits or other pieces inside the miniDOT logger (if the sensor is not completely dry).
2. Attach the logger to a Windows or Mac computer that has Java installed via the provided micro-USB cable. The miniDOT logger will appear as a Drive on the computer. If there are connection issues or the LED is flashing red, refer to the manufacturer's manual.
3. Run the miniDOTControl.jar program to see the state of the miniDOT Logger. (i.e. sensor calibration date, current logger time, sample interval, battery voltage, temperature, etc.)
 - a. After opening the program, click the "Connect" button to activate the connection between the miniDOT logger and computer. The different parameters displayed will populate with the information from the logger. When this software is launched it will synchronize the clock of the sensor with the clock of the computer and report if the clock onboard the logger has been adjusted.
 - b. To adjust the recording time interval, click the "Set Sample Interval" button. Then modify the sampling time to any time greater than 1 minute and less than 60 minutes. Then click the "Set Sample Interval" button again.
 - i. Suggested interval for the GEM program is 1 minute.
4. When finished with configuration, simply close the software window and disconnect the usb cable. If initiating logging immediately, the logger control switch can be flipped to "record" and the pressure housing reinstalled. Ensure the desiccant pack is in place and that the o-ring and seat are clear of debris. If initiating logging later simply close up the unit, and reopen later when it is time to initiate logging.

Data Transfer to a Computer:

1. Wipe off the exterior white pressure housing with a cloth to ensure no water enters the PME miniDOT logger in the following steps.
2. Open the miniDOT logger by unscrewing the white pressure housing from the black end cap. Remove the white pressure housing completely, being careful not to drip water onto the interior surfaces of circuits or other pieces inside the miniDOT logger (if the sensor is not completely dry).

3. If still logging, turn the logger switch from the “Record” position to the “Halt” position. The logger must be halted when connected to the computer.
4. Attach the logger to the computer via the micro-USB cable supplied.
5. Copy the files from the folder with the same serial number as the miniDOT logger to the computer. The minidot will generate a new file every 24 hours if in continuous operation, or it will generate a new file every time logging is stopped and restarted. The files are named by the time of the first measurement within the file (YYYY-MM-DD HHMMSSZ.txt)
6. Optional - Run the miniDOTPlot.jar program to see a plot of the measurements
 - a. After opening the program, enter an elevation of 0m or a barometric pressure of 1013.25 mbar and a salinity value of 0 PPT. We do not input a Salinity value at this stage for simplicity - it is likely that there will be multiple water depth samples in the same file which have different associated salinities which could cause confusion and potential inappropriate interpretation of erroneous data. Putting in 0 ensures that only raw data is being plotted.
 - b. Click “Select Data Folder” and select the appropriate folder for the file of interest.
 - c. Select “Plot” to plot the data measurements.

The plot properties can be altered and saved with ‘right click’ on the plot that is displayed. Remember that the oxygen data shown is uncorrected for salinity and pressure effects.
5. Optional - Run the miniDOTConcatenate.jar program to combine multiple files into one CAT .txt file, produce a ‘yyyy-mm-dd hh:mm:ss’ timestamp and calculate % saturation (with caveats)
 - a. After opening the program, enter a barometric pressure of 1013.25 mbar or elevation of 0m at sea level and a salinity measurement of 0 PPT. If you deviate from these inputs the raw DO mg/L will be altered. Since we are uploading raw data to the repository at this stage, we will not explore salinity corrections at this stage.
 - b. Click the “Concatenate” button to create a single CAT .txt file.
 - c. The CAT .txt file is useful for preliminary data exploration and completing the instrument data processing sheet but it is not necessary. This can all be done with the raw files. The raw files are what are submitted to the repository.
6. Once data has been downloaded, delete the data in the measurement folder of the miniDOT logger, but be careful not to delete the miniDOTControl program or the other .jar programs.
7. Disconnect the minDOT logger from the USB port on the computer.

8. Make sure there is no debris on the o-ring seal before closing the miniDOT logger. Screw the white pressure housing back onto the black end cap. Install the storage cap with or without a wet kimwipe or sponge depending on intended storage length.

Hach DR1900

Data Transfer to a Computer (instrument can store up to 500 data points internally):

1. Remove the connector cover piece and insert the USB+Power module into the slot on the underside of the DR1900.
 - a. Note: Do not connect the USB+Power module to the computer when the module is still connected to the DR1900 instrument.
2. Push the < SETTINGS > button on the DR1900 instrument (the wrench symbol).
3. Scroll down to select the 'Data Management' option.
4. Scroll down to select the 'Send Data' option. All data that is stored on the DR1900 will be sent to the USB+Power module as a .TXT file.
 - a. The screen will read "Sending Data" while being read to the module and will display "Transfer Complete" when the data is successfully transferred to the module.
 - b. Exit the data transfer screen and turn off the DR1900 instrument.
5. Remove the USB+Power module from the DR1900 instrument and reinstall the connector cover piece on the underside of the instrument.
6. Using the USB to type A cable, plug the type A connector into the module and the USB connector into the computer.
7. Open the DR1900 USB module on the computer.
 - a. It will show up on the computer as a USB storage device and will most likely have the name '2100Q USB' associated with it.
8. The data recently sent to the USB module may be found in the folder 'datalog' or in the main directory.
 - a. The filename will be formatted as: LOG-YYMMDDHHMMSS.TXT
9. Copy and paste the data file to a location on the computer.
10. Open a spreadsheet program, like Excel, to view the data.
 - a. If opening with Excel, the datafile will be comma delimited so the user will have to format the 'text to columns'.

Turner Designs AquaFluor

Data Transfer to a Computer (instrument can store up to 1000 data points internally):

1. Connect the AquaFluor to the serial port of a computer.
2. Open the Turner Designs Spreadsheet Interface Software.
3. Press <DATA> button 3 times.
4. Press <ENT> button 5 times to start the data download.
5. When finished, press <ESC> to return to the Home screen.

To erase stored data after download:

6. Press <DATA> button 4 times.
7. Press <ENT> button 5 times to erase all logged data.
8. When finished, press <ESC> to return to the Home screen.